# Venkata Lakshmi Parimala Pasupuleti

Senior Engineer · Multi-Agent AI Systems · LLM Production Engineering · RAG Pipelines · LangGraph · Distributed Systems · Python · Docker · Kubernetes · AWS · NLP · CI/CD

+1-703-261-5410 | lakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

## PROFESSIONAL SUMMARY

Senior Python engineer with 8 years building large-scale distributed systems and two shipped multi-agent AI applications in production - directly matching GEICO's Service Engineering mission. LeaseI is an 8-agent LangGraph stateful system enabling intelligent, collaborative decision-making: multi-source RAG, NLP-driven natural language understanding, HITL escalation, and deterministic guardrails for regulated decisions - 93.6% precision across 1,240 compliance test cases. AgentProbe is a production LLM evaluation platform with CI-integrated regression gating, multi-provider LLM orchestration, and observability across AI quality metrics. Entrata background brings distributed systems at scale: RabbitMQ async processing, Redis caching, Docker containerization (50% scalability improvement), AWS S3/EC2/RDS, and 99.5% uptime across 100K+ monthly active users. NLP experience spans BERT embeddings, spaCy, and LLM-powered natural language query systems (AI on FHIR). M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

## TECHNICAL SKILLS

**AI & Multi-Agent Systems:** Multi-Agent Orchestration, LangGraph, LangChain, Agentic AI, RAG Pipelines, LLM Orchestration, LLM Evaluation, ChromaDB, Vector Search, HITL, Hallucination Detection, Guardrails, Prompt Engineering, OpenAI API, Anthropic API, Groq, Ollama, BERT, spaCy, NLP, PyTorch, scikit-learn, MCP

**Engineering & Distributed Systems:** Python (8yr), JavaScript, TypeScript, SQL, NoSQL, FastAPI, Flask, Docker, Kubernetes, Microservices, REST APIs, CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue), Azure-compatible cloud architecture, RabbitMQ, Redis, PostgreSQL, MySQL, Oracle, Bash, Linux, Git

**Reliability & Observability:** Application Monitoring, Performance Assessment, Production Reliability, Observability, Distributed System Design, Scalable Architecture, Fault-Tolerant Systems, PHI Compliance, Feature Engineering, NumPy, Pandas, PySpark, Tableau, Agile/Scrum

## EXPERIENCE

### Magizhs Tech Solution

May 2025 - Aug 2025

**AI/ML Engineer Intern**

- Designed and deployed a large-scale distributed AI application (LangGraph, Python) for mission planning - multi-agent orchestration with NLP-driven natural language understanding, RAG retrieval over distributed sources, and deterministic validation guardrails before delivery; CI/CD-integrated automated testing at every release.
- Implemented LLM-powered multi-step workflows with observability on agent performance, retrieval quality, and latency; HITL escalation for edge cases requiring human review.

### George Mason University

Aug 2024 - May 2026

**Graduate Teaching Assistant & Research Associate**

- Built Python ML pipelines (PyTorch, scikit-learn) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end from data ingestion through model evaluation and statistical validation across 190+ countries.
- Mentored 600+ students in Python, SQL, and distributed systems; designed monitoring dashboards improving research comprehension by 40%.

### Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

**Senior Software Engineer · Lehi, UT (Remote)**

- **Designed and deployed large-scale distributed systems at production scale** - 20+ scalable microservice modules at 99.5% uptime across 100K+ monthly active users; Docker containerization (50% scalability improvement, 30% deployment reduction); AWS S3, EC2, RDS at enterprise scale; fault-tolerant async processing (RabbitMQ) with retry, idempotency, and dead-letter guarantees.
- **Integrated AI/ML components into production platform** - AI-driven observability and monitoring reducing triage time 40%; ML-powered data validation across compliance-critical workflows; Python-based automation pipelines eliminating 80% of manual operational workload.
- **Performance assessment, monitoring, and reliability engineering** - internal application monitoring tooling; 200+ SQL/PostgreSQL/Oracle query optimizations (30% performance gain) through execution plan analysis, indexing, and partitioning; maintained sub-10 bug count for 6+ consecutive months across multi-tenant production environments.
- **Secure, scalable distributed system design** - 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents); Redis caching; REST API development across regulated SaaS platform; continuous delivery with GitHub Actions CI/CD; 300+ code reviews; mentored 5+ engineers on distributed systems and AI best practices.

### Entrata India - Pune

Jan 2017 - Sep 2018

**Associate Software Engineer → Trainee**

- Built distributed backend modules (PHP, Python, PostgreSQL, NoSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

## PROJECTS

### Leasel: 8-Agent Multi-Agent AI System

Multi-Agent · RAG

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_raq

Large-scale distributed AI application with intelligent multi-agent orchestration: 8-agent LangGraph stateful pipeline enabling collaborative decision-making, multi-source RAG with NLP-driven retrieval, HITL escalation when model confidence is below threshold, and deterministic guardrails for regulated outputs. 93.6% precision, 87.9% F1 across 1,240 test cases; CI quality gate with full observability.

### AgentProbe: Production LLM Evaluation & Observability Platform

LLM Eval · Monitoring

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Production AI monitoring and observability platform: 15-endpoint FastAPI service, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated regression gate, LLM-as-judge scoring, multi-provider LLM routing for cost/performance optimization. Application monitoring across 1,240+ test cases - zero regressions shipped to production.

### AI on FHIR: NLP-Driven Natural Language Query System

NLP · Distributed

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Distributed AI application translating natural-language queries into FHIR API calls via BERT embeddings and LLM inference - NLP-powered customer self-service pattern reducing manual retrieval time ~60%; HIPAA-compliant RBAC with full data isolation across structured records.

## EDUCATION

**M.S. Data Analytics Engineering -** George Mason University, Virginia — May 2026 · GPA 3.87

**B.Tech Computer Science & Engineering -** JNTU Anantapur — 2013-2017